

A SIMPLE PROOF OF $P \neq NP$

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A proof of the $P \neq NP$ problem such that a grade schooler should be able to understand. I am writing for my daughters Margo (10) and Nora (12).

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*Dedicated to the memory of those killed at Virginia Tech,
and in solidarity with those who were injured, their families,
and the Virginia Tech community.*

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DEFINITIONS

Before going into the proof we need to clarify what the complexity classes **P** and **NP** are.

REFERENCES

1. Stephen Cook, “The complexity of theorem-proving procedures,” *Conference Record of Third Annual ACM Symposium on Theory of Computing*, pages 151-158, 1971.
2. Stephen Cook, “The **P** versus **NP** Problem,” *Official Problem Description for the Millennium Prize Problems*, Clay Mathematics Institute, 2000, <https://www.claymath.org/wp-content/uploads/2022/06/pvsnp.pdf>